

CompositionMeasure for MaxDivergence

Michael Shoemate (@Shoeboxam)

This proof resides in “**contrib**” because it has not completed the vetting process.

Proves soundness of the implementation of **CompositionMeasure** for **MaxDivergence** in **mod.rs** at commit **f5bb719** (outdated¹).

1 Hoare Triple

Precondition

Compiler-verified

Types matching pseudocode.

Caller-verified

None

Pseudocode

```
1 class CompositionMeasure(MaxDivergence):
2     def composability( #
3         self, adaptivity: Adaptivity
4     ) -> Composability:
5         return Composability.Concurrent
6
7     def compose(self, d_mids: Vec[tuple[Self.Distance, u32]]) -> Self.Distance:
8         d_out = 0.0
9         for d_mid, k_i in d_mids:
10             d_out = d_out.inf_add(d_mid.inf_mul(f64.from_(k_i)))
11         return d_out
```

Postcondition

Theorem 1.1. `composability` returns `Ok(out)` if the composition of a vector of privacy parameters `d_mids`, where each `d_mid` is charged `k` times, is bounded above by `self.compose(d_mids)` under `adaptivity` and `out-composability`. Otherwise returns an error.

Proof. By the postconditions of **InfAdd** and **InfMul** we have that $\sum_i k_i d_mids_i \leq \text{compose}(d_mids)$.

Adaptivity	Sequential	Concurrent
Non-Adaptive	Theorem 1[DMNS06]	Theorem 1.8[VW21]
Adaptive	Theorem 1[DMNS06]	Theorem 1.8[VW21]
Fully-Adaptive	None	None

This table is reflected in the implementation of `composability` on line 2.

¹See new changes with `git diff f5bb719..d7b4364b rust/src/combinators/sequential_composition/mod.rs`

□

References

- [DMNS06] Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. Calibrating noise to sensitivity in private data analysis. In Shai Halevi and Tal Rabin, editors, *Theory of Cryptography*, pages 265–284, Berlin, Heidelberg, 2006. Springer Berlin Heidelberg.
- [VW21] Salil Vadhan and Tianhao Wang. Concurrent composition of differential privacy, 2021.